Projection maps construction
Project 2

To begin, we will discuss some preliminary facts about our spaces. In the following, we will discuss about triangulations of a mesh $\mathcal{T}_h$ in $\mathbb{R}^3$ in this case these are tetrahedron $K \in \mathcal{T}_h$. In the following definitions $x = [x_1, x_2, x_3]^T$. Note that these are finite dimensional function spaces.

$$\begin{aligned} V_h^0(K) =& \{v : v = a + b \cdot x, \text{ with } a \in \mathbb{R}, b \in \mathbb{R}^3\}, \\ V_h^1(K) =& \{v : v = a + b \times x, \text{ with } a \in \mathbb{R}^3, b \in \mathbb{R}^3\}, \\ V_h^2(K) =& \{v : v = a + bx \text{ with } a \in \mathbb{R}^3, b \in \mathbb{R}\}, \\ V_h^3(K) =& \{v : v = a \text{ with } a \in \mathbb{R}\}. \end{aligned}$$

With these function spaces we can discuss how to uniquely determine these functions through their degrees of freedom.

Degrees of freedom for $V_h^0(K)$

$$v(y), \text{ for all vertices } y \text{ of } K. \tag{0.1}$$

Note that since the tetrahedra K has four vertices we have exactly four degrees of freedom. Notice that the dimension of the space $V_h^0(K)$ is exactly 4 and that a function $v \in V_h^0(K)$ is uniquely determined by its degrees of freedom. In general this would be total number of vertices.

Degrees of freedom for $V_h^1(K)$

$$\int_e v \cdot t, \text{ for all edges } e \text{ of } K. \tag{0.2}$$

Here t denotes a unit tangent vector to e. Since K has six edges, there are six degrees of freedom. The dimension of $V_h^1(K)$ is six as well.

Degrees of freedom for $V_h^2(K)$

$$\int_f v \cdot n, \qquad \text{for all faces } f \text{ of } K. \tag{0.3}$$

Since K has four faces there are four degrees of freedom. The dimension of $V_h^2(K)$ is four as well.

Degrees of freedom for $V_h^3(K)$

$$\int_K v. \tag{0.4}$$

The dimension of $V_h^3(K)$ is the number of tetrahedra K, which is 1, and is exactly our degrees of freedom.

We now perform a gluing operation to connect these piecewise linear functions to one space continuously. These can be shown to be equivalent to the degrees of freedom matching on their

respective V_h^k simplex, This allows us to check that these are in fact continuous functions even through multiple tetrahedra.

$$V_h^k = \{v : v|_K \in V_h^k(K), \text{ for all } K \in \mathcal{T}_h, \text{ the corresponding d.o.f are single-valued}\} \tag{0.5}$$

This means for where the tetrahedron meet the degrees of freedom that coincide with our V_h^k space must agree. For example, when 2 tetrahedron meet, 3 vertices meet at the face. Hence the 3 values must be the same for both functions at the vertices for V_h^0. Similarly, we do the same for the edges for V_h^1, then the faces for V_h^2, then whole tetrahedron integral over v for for V_h^3.

1. Projection maps

Now we can define a projection map on these spaces

$$\|u\|^2_{H(\text{grad},\Omega)} = \|u\|^2_{L^2(\Omega)} + \|\text{grad} u\|^2_{L^2(\Omega)} \tag{1.1}$$

$$\|u\|^2_{H(\text{curl},\Omega)} = \|u\|^2_{L^2(\Omega)} + \|\text{curl}\, u\|^2_{L^2(\Omega)} \tag{1.2}$$

$$\|u\|^2_{H(\text{div},\Omega)} = \|u\|^2_{L^2(\Omega)} + \|\text{div}\, u\|^2_{L^2(\Omega)}. \tag{1.3}$$

We define the corresponding spaces as follows.

$$H(\text{grad},\Omega) := \{u : \|u\|_{H(\text{grad},\Omega)} < \infty\} \tag{1.4}$$

$$H(\text{curl},\Omega) := \{u : \|u\|_{H(\text{curl},\Omega)} < \infty\} \tag{1.5}$$

$$H(\text{div},\Omega) := \{u : \|u\|_{H(\text{div},\Omega)} < \infty\}. \tag{1.6}$$

Let us use the following notation.

$$W^0 = L^2(\Omega),\ W^1 = [L^2(\Omega)]^3,\ W^2 = [L^2(\Omega)]^3,\ W^3 = L^2(\Omega)$$

and

$$V^0 = H(\text{grad},\Omega),\ V^1 = H(\text{curl},\Omega),\ V^2 = H(\text{curl},\Omega),\ V^3 = L^2(\Omega)$$

we want these weight functions to act linearly on the chains.

$$Z^0(\sum_y a_y y) = \sum_y a_y Z^0(y)$$

Similarly we can define Z^k on k-chains.

We then define

$$\Pi_h^0 v(y) = \int_\Omega (Z^0(y))(x) v(x) dx \qquad \forall \text{vertices } y$$

$$\int_e \Pi_h^1 v \cdot t_e = \int_\Omega (Z^1(e))(x) \cdot v(x) dx \qquad \forall \text{edges } e$$

$$\int_F \Pi_h^2 v \cdot n_F = \int_\Omega (Z^2(F))(x) \cdot v(x) dx \qquad \forall \text{faces } F,$$

$$\int_K \Pi_h^3 v = \int_\Omega (Z^3(K))(x) v(x) dx \qquad \forall \text{tetrahedra } K$$

and we want them to commute hence we seek operators with $\Pi_h^k : W^k \to V_h^k$, $k = 0, 1, 2, 3$ such that

$$\Pi_h^k v = v \qquad \text{if } v \in V_h^k \tag{1.7}$$

$$\operatorname{grad} \Pi_h^0 v = \Pi_h^1 \operatorname{grad} v \qquad \text{if } v \in V^0 \tag{1.8}$$

$$\operatorname{curl} \Pi_h^1 v = \Pi_h^2 \operatorname{curl} v \qquad \text{if } v \in V^1 \tag{1.9}$$

$$\operatorname{div} \Pi_h^2 v = \Pi_h^3 \operatorname{div} v \qquad \text{if } v \in V^2 \tag{1.10}$$

$$\|\Pi_h^k v\|_{L^2(\Omega} \leq C_k \|v\|_{L^2(\Omega)} \qquad \text{for if } v \in W^k \tag{1.11}$$

these conditions are equalent to these properties

LEMMA 1. *If the weights satisfy the following properties*

$$\begin{aligned} -div\, Z^1(e) &= Z^0(\partial_1 e) \\ curl\, Z^2(F) &= Z^1(\partial_2 F) \\ -grad\, Z^3(K) &= Z^2(\partial_3 K) \end{aligned}$$

then, the projections satisfy the commuting properties, (1.8),(1.9),(1.10).

To maintain that these are projections we need the following lemma.

LEMMA 2. *Suppose that*

$$\begin{aligned} \int_\Omega (Z^0(y))(x) v(x) dx &= v(y) \qquad \forall v \in V_h^0, \forall \text{vertices } y \\ \int_\Omega (Z^1(e))(x) \cdot v(x) dx &= \int_e v \cdot t_e \qquad \forall v \in V_h^1, \forall \text{edges } e \\ \int_\Omega (Z^2(F))(x) \cdot v(x) dx &= \int_F v \cdot n_F \qquad \forall v \in V_h^2, \forall \text{faces } F, \\ \int_\Omega (Z^3(K))(x) v(x) dx &= \int_K v \qquad \forall v \in V_h^3, \forall \text{tetrahedra } K \end{aligned}$$

then, (1.7) *holds*

Our goal is to solve for $Z^0(y)$. By obtaining our Z^0 weight, we can find the Z^1, Z^2 and Z^3 weights. **Lilly:** Our goal for our meeting 07-08: to get psi and the first weight Z^0

Our first weight Z^0 can be defined by this formula:

$$Z^0(y) = \eta_0(y) - \operatorname{div}(\mu_y \operatorname{grad} \psi_0(y))$$

To use this formula we require some preliminary definitions.

Definition 1. *For a vertex y we define the star of y as follows*

$$st(y) = \bigcup_{y \in K} K$$

where the union is taking over the tetrahedra K.

Definition 2. $V_h^0(st(y)) = \{v|_{st(y)} : v \in V_h^0\}$

$V_h^0(st(y))$ is the set of all functions obtained by taking a function from V_h^0 and looking only at its values on the star of the vertex y.

Definition 3. *From our original mesh $\mathcal{T}_h$ we define the Alfeld mesh $\mathcal{T}_h^A$ by choosing the barycenter, denoted x_K, for each $K \in \mathcal{T}_h$ connecting to all the vertices of K in a straight line.*

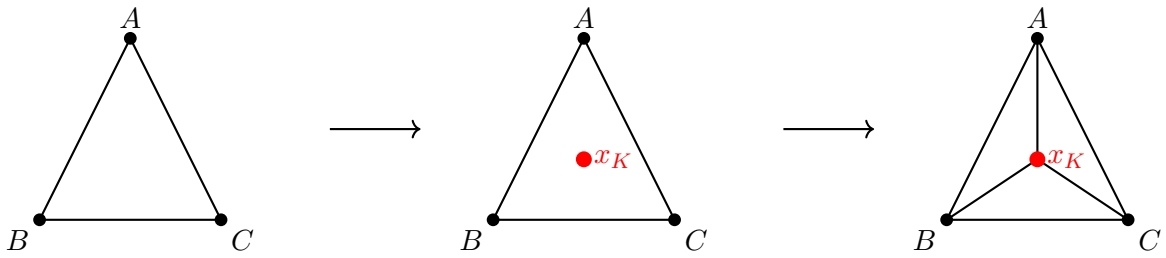


FIGURE 1. Construction of the Alfeld split. Starting from a simplex K, its barycenter x_K is introduced and connected to each vertex, subdividing K into smaller simplices. Repeating this process for every element of the mesh $\mathcal{T}_h$ produces the Alfeld mesh $\mathcal{T}_h^A$.

The corresponding FEEC spaces on the new mesh $\mathcal{T}_h^A$ will be denoted by $V_{h,A}^k$.

The barycenter can be thought of as a center of mass. We perform the Alfeld split because it creates more elements. By creating more elements we can build our finite element spaces with stronger continuity. Essentially by changing the mesh, we make the finite element construction work better. Before the Alfeld split we were working in V_h^k, after refining the mesh with the Alfeld split, we build the same type of finite element space, but on a new mesh $V_{h,A}^k$.

We introduce the indicator function now, for the definitions that follow shortly after it.

Definition 4. *Definition of indicator function on general set A*

$$\chi_A(x) = \begin{cases} 1 & \text{if } x \in A, \\ 0 & \text{if } x \notin A. \end{cases}$$

Definition 5. *Let $\mu \in V_{h,A}^0$ be such that μ vanishes at all the vertices of the original mesh $\mathcal{T}_h$ and takes value one on the barycenter x_K for all $K \in \mathcal{T}_h$. For a vertex y of $\mathcal{T}_h$ we let*

$$\mu_y = \mu\chi_{st(y)}$$

where $\chi_{st(y)}$ is the indicator function of $st(y)$.

Definition 6.

$$\eta_0(y) = \frac{1}{|st(y)|}\chi_{st(y)}$$

where $\frac{1}{|st(y)|} = \int_{st(y)} 1dx$ indicates the volume of $st(y)$. Thus,

$$(\eta_0(y))(x) = \begin{cases} \frac{1}{|st(y)|} & \text{if } x \in st(y) \\ 0 & \text{if } x \notin st(y) \end{cases}$$

Definition 7. $\mathfrak{Z}V_h^0 = \{v \in V_h^0(st(y)) : gradv = 0\}$

Note that $\mathfrak{Z}V_h^0 = \{v \in V_h^0(st(y)) : \text{grad}v = 0\} = ker(\text{grad}, V_h^0(st(y))$

Definition 8.

$$\mathfrak{Z}^{\perp}V_h^0(st(y)) = \{v \in V_h^0 : \langle v, u\rangle = 0, \forall u \in \mathfrak{Z}V_h^0\}$$

The space $\mathfrak{Z}^{\perp}V_h^0(\text{st}(y))$ is called the *orthogonal complement* of the kernel. It consists of all functions in $V_h^0(\text{st}(y))$ that are orthogonal to every function in $\mathfrak{Z}V_h^0$ with respect to the chosen inner product. Two functions u and v are orthogonal if

$$\langle u, v\rangle = 0.$$

For the L^2 inner product, this means

$$\int_{\text{st}(y)} uv\, dx = 0.$$

The zero function is trivially orthogonal to every function, but there are also many nonzero functions that satisfy this condition. Thus,

$$\mathfrak{Z}^{\perp}(\text{st}(y)) = \{v \in V_h^0 st(y) : \int_{st(y)} v(x)dx = 0\}$$

Now, we just need to specify $\psi_0(y)$. It is not given implicitly, it has to be solved using a linear system.

The function $\psi_0(y) \in \mathfrak{Z}^{\perp}V_h^0(\text{st}(y))$ satisfies

$$\langle \mu_y \text{grad}\psi_0(y), \text{grad}u\rangle_{\text{st}(y)} = u(y) - \langle \eta_0(y), u\rangle_{\text{st}(y)} \qquad \forall u \in \mathfrak{Z}^{\perp}V_h^0(\text{st}(y))$$

Through this we can solve for $\psi_0(y)$.

Now that we know the preliminary definitions, specifically $\mu_y, \eta_0(y), \psi_0(y)$ we can solve for Z^0

$$Z^0(y) = \eta_0(y) - \text{div}(\mu_y \text{grad}\psi_0(y))$$

With this, we hope we can construct our weight function hence our first projection.